

Bank Loan Risk Management Project - 10 SQL Queries

1. How many loan applications are in the dataset?

```
SELECT COUNT(*) AS Total_Customers FROM Loan_Customers;
```

2. What is the distribution of loan status (Paid vs Default)?

```
SELECT Loan_Status, COUNT(*) AS Total_Customers FROM Loan_Customers GROUP BY Loan_Status;
```

3. What is the average loan amount for each loan purpose?

```
SELECT Loan_Purpose, ROUND(AVG(Loan_Amount),2) AS Avg_Loan_Amount FROM Loan_Customers GROUP BY Loan_Purpose ORDER BY Avg_Loan_Amount DESC;
```

4. Which regions have the highest number of loan defaults?

```
SELECT Region, COUNT(*) AS Default_Count FROM Loan_Customers WHERE Loan_Status='Default' GROUP BY Region ORDER BY Default_Count DESC;
```

5. What is the average credit score of customers who defaulted?

```
SELECT ROUND(AVG(Credit_Score),2) AS Avg_Default_Credit_Score FROM Loan_Customers WHERE Loan_Status='Default';
```

6. Which employment status has the highest default rate?

```
SELECT Employment_Status, COUNT(*) AS Default_Customers FROM Loan_Customers WHERE Loan_Status='Default' GROUP BY Employment_Status ORDER BY Default_Customers DESC;
```

7. What is the average debt-to-income ratio by loan status?

```
SELECT Loan_Status, ROUND(AVG(Debt_to_Income_Ratio),2) AS Avg_DTI FROM Loan_Customers GROUP BY Loan_Status;
```

8. Find the top 10 customers with the highest loan amounts.

```
SELECT Customer_ID, Loan_Amount, Credit_Score FROM Loan_Customers ORDER BY Loan_Amount DESC LIMIT 10;
```

9. How many customers have a credit score below 600?

```
SELECT COUNT(*) AS High_Risk_Customers FROM Loan_Customers WHERE Credit_Score < 600;
```

10. Which occupations have the highest average loan amount?

```
SELECT Occupation, ROUND(AVG(Loan_Amount),2) AS Avg_Loan_Amount FROM  
Loan_Customers GROUP BY Occupation ORDER BY Avg_Loan_Amount DESC;
```